THE COMPOUNDING BOTTLENECK: A UNIFIED CAPACITY ACCOUNT OF RETRIEVE-THEN-COMPRESS RETRIEVAL-AUGMENTED GENERATION

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ABSTRACT

Efficient retrieval-augmented generation (RAG) funds two fixed-dimensional vector bottlenecks out of one budget: the dense-retrieval embedding of dimension d_r, and the soft-context compression code of size $m \cdot d_c$. Prior work bounds the retrieval stage or studies compression overflow, but never the composition. We give a unified capacity account that treats both stages as thresholded-separation problems of the same geometric form, and we make four observations, each validated from controlled free-vector geometry up to production encoders. First, the two stages carry sharp capacity walls whose critical dimension grows with content; the retrieval wall replicates across two embedder families, two datasets, and optimal (PCA) truncation, and on the adversarial LIMIT benchmark three real embedders collapse to 2.8–8.2% recall@100. Second, composing the stages *compounds*: the best split of a shared budget underperforms either stage given the full budget, and the sign of the deviation from the product law is governed by the correlation ρ between which items are hard to retrieve and which are hard to compress. Third, we measure ρ on a real pipeline and find $\rho \approx 0$, so real RAG sits in the multiplicative regime and a simple product-law allocation is near-exact; the allocation is interior, which we confirm both on a synthetic corpus and on a real multi-hop QA benchmark with a frozen reader scored by EM/F1 (a single-peaked allocation curve and a 0.048 F1 compounding gap at a tight budget, over three seeds), and deployed big-embedder/tiny-compressor configurations sit far off the frontier. Fourth, a multi-view "facet-lens" escapes the wall in the free-vector worst case, but we show honestly that the escape does not transfer to strong real encoders, which redirects the practical lever back to allocation. All claims are reproducible from committed configurations.

1 INTRODUCTION

Retrieval-augmented generation has settled into a common efficient template: a query is embedded and matched against a corpus by a dense retriever (Karpukhin et al., 2020; Lewis et al., 2020), the top passages are compressed into a handful of soft tokens so the generator pays little attention cost (Mu et al., 2023; Ge et al., 2023; Cheng et al., 2024), and a frozen decoder answers from those tokens. Both stages are, at heart, a fixed-dimensional vector bottleneck: retrieval squeezes relevance through an embedding of dimension d_r, and compression squeezes a passage through a code of size $m \cdot d_c$. The two bottlenecks are usually studied apart. On the retrieval side, a line of work now argues that a d_r-dimensional single-vector index cannot realize arbitrary top-k relevance patterns, and Weller et al. (2025) make this precise by tying feasibility to the sign-rank of the relevance matrix and by building the LIMIT stress test on which frontier models fail. On the compression side, Belikova et al. (2026) observe empirically that soft tokens "overflow" past a finite capacity, and Liu et al. (2026) argue on information-theoretic grounds that full compression is neither feasible nor necessary, yet decline to give a capacity bound or to connect compression back to the embedding-dimension wall.

What has been missing is the composition. Efficient RAG runs the compressor on the *output* of the retriever, so the two capacity-limited stages act in series, and the natural question is whether their

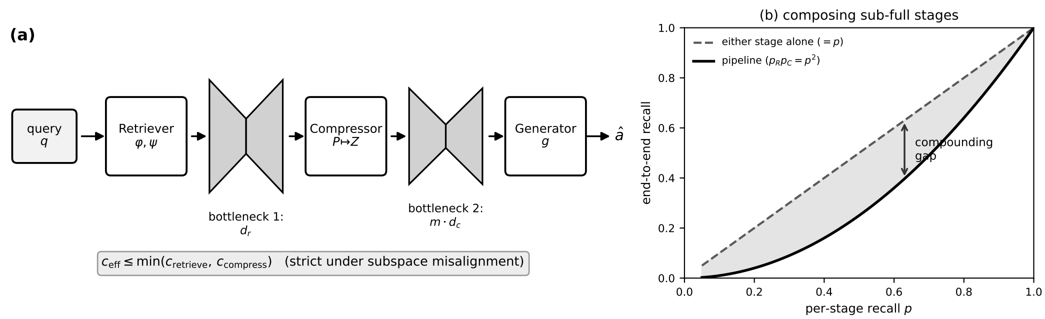

Figure 1: **The compounding bottleneck.** (a) Efficient RAG passes a query through two fixed-dimensional vector bottlenecks in series—the retrieval embedding (d_r) and the soft compression code ($m \cdot d_c$)—so the realizable set is the intersection of two subspace-constrained families and $c_{\text{eff}} \leq \min(c_{\text{retrieve}}, c_{\text{compress}})$, strict under generic misalignment. (b) Composing two stages that each achieve recall $p < 1$ yields $p_R p_C$ under independence, which lies below either stage alone; the shaded region is the compounding gap.

errors simply add, or whether stacking two bottlenecks does something worse. We find that they *compound*: there are inputs on which retrieval alone succeeds and on which compressing the gold passage alone succeeds, and yet the composed pipeline fails, and the size of this effect is predictable from the geometry of the two representations rather than from either stage's standalone metric. This breaks the field's implicit assumption that retrieval error and compression error are independent and additive, and it has a direct practical consequence, since a budget spent overwhelmingly on a large embedding while the compressor is starved sits far from the optimal allocation.

We organize the paper around four contributions, illustrated in Figure 1. (C1) We cast both stages as thresholded-separation problems of the same geometric form and show each has a sharp capacity wall whose critical dimension grows with content; the retrieval wall is not an artifact of one embedder, one dataset, or one truncation scheme, and it appears in full force on Weller's own adversarial data. (C2) We show that composing two sub-full stages compounds, we characterize the deviation from the product law through a hardness-correlation ρ, and—going beyond the prior framing that left ρ hypothetical—we *measure* it on a real pipeline and find it near zero, which places real RAG squarely in the multiplicative regime. (C3) Under a fixed shared budget we derive and validate the optimal split, on synthetic corpora and end-to-end on real encoders, and show that deployed configurations are off the frontier. (C4) We test the obvious escape—a multi-view facet-lens that gives each semantic facet its own low-rank subspace—and report the honest result that it escapes the wall in the free-vector worst case but *not* on strong real encoders, which we explain and which sharpens rather than weakens the account. Every headline number is validated twice, from a free-vector best case that upper-bounds any encoder to production embedders, and the negatives are reported alongside the positives because they are what map the boundary of the effect.

2 Related work

Limits of dense retrieval. That a fixed embedding dimension caps what a single-vector retriever can express is the premise we build on. Weller et al. (2025) connect the number of realizable top-k document subsets to embedding dimension through sign-rank (Razborov & Sherstov, 2010; Forster, 2002) and release LIMIT, where even strong models score low; scaling studies of embedding dimension in IR and analyses of granularity failures point the same way. Matryoshka representation learning (Kusupati et al., 2022) makes the leading coordinates of an embedding independently useful, which is exactly what lets us trace the wall by truncation on a real encoder. Multi-vector late interaction (Khattab & Zaharia, 2020; Santhanam et al., 2022) is the known partial escape from the single-vector bound, and we position our facet-lens against it. Standard dense retrievers and embedders we use or reference include Reimers & Gurevych (2019); Karpukhin et al. (2020); Izacard et al. (2022); Wang et al. (2022); Li et al. (2023b); Xiao et al. (2023); Merrick et al. (2024); Ni et al. (2022), benchmarked in the style of Thakur et al. (2021); Muennighoff et al. (2023).

Context and soft-prompt compression. Soft-compression methods replace a passage with a few learned vectors read by a frozen decoder: Gist tokens (Mu et al., 2023), in-context autoencoding (Ge et al., 2023), context-adapted models (Chevalier et al., 2023), and one-token extreme compression (Cheng et al., 2024), alongside hard-prompt pruning (Jiang et al., 2023a;b; Pan et al., 2024; Li et al., 2023a) and abstractive compression (Xu et al., 2024; Li et al., 2024). Closest to us, Belikova et al. (2026) detect an "overflow" regime where compressed tokens can no longer answer a query, and Liu et al. (2026) argue that query-conditioned selection beats full compression; both stop short of a capacity law and of any link to the retrieval wall. We supply that link and, crucially, the cross-stage composition.

RAG systems. The broader pipeline we abstract is instantiated by Lewis et al. (2020); Guu et al. (2020); Borgeaud et al. (2022); Izacard et al. (2023); Izacard & Grave (2021); Ram et al. (2023); Shi et al. (2024); Asai et al. (2024). Our contribution is orthogonal: not a better pipeline, but a capacity account of why stacking two fixed-dimensional codes loses information that either stage alone would keep, and what to do about it.

3 Setup and the capacity primitive

Notation. A corpus is $D = \{d_1, \ldots, d_N\}$ with queries q and a ground-truth relevance matrix $A \in \{0,1\}^{|Q| \times N}$, where row i marks the documents relevant to query i. The retrieval stage uses a document encoder $\varphi : d \mapsto x \in \mathbb{R}^{d_r}$ and a query encoder $\psi : q \mapsto y \in \mathbb{R}^{d_r}$, scores by $s(q,d) = \langle y, x \rangle$, and returns the top k. Realizing A by thresholding $YX^\top$ is governed by the sign-rank of $2A - 1$ (Weller et al., 2025), which is the retrieval wall. The compression stage maps a passage P to $Z \in \mathbb{R}^{m \times d_c}$ (m soft tokens of width d_c) that a frozen reader answers from; we cast "can the reader recover the queried fact from Z among distractors" as a second thresholded-separation problem with its own capacity c_{compress}. Because the compressor acts on the retriever's output, the two capacities compose by set intersection, which we state as follows and validate empirically throughout the paper.

Proposition 1 (Compositional capacity)**.** *Let $\mathcal{R}_r$ be the family of relevance/answer patterns realizable by the retrieval stage at capacity c_{retrieve}, and $\mathcal{R}_c$ the family realizable by the compression stage at c_{compress}. A pattern is realized end-to-end only if the retriever surfaces the gold set* and *the compressor preserves the answer from it, so the pipeline's realizable family is $\mathcal{R}_r \cap \mathcal{R}_c$ and its effective capacity satisfies $c_{\text{eff}} = |\mathcal{R}_r \cap \mathcal{R}_c| \leq \min(c_{\text{retrieve}}, c_{\text{compress}})$. Equality holds iff one family contains the other; generically it does not—the retriever's* discriminative *subspace and the compressor's* reconstructive *subspace are misaligned— so the inequality is strict, and the shortfall is the failure-transfer set we measure in Section 5.*

The proof is immediate (a pattern in neither stage's realizable family cannot be realized by their composition); the content is empirical, namely *how far* below the minimum the composition falls, which is exactly the compounding gap C2 quantifies.

The free-embedding capacity measure. To measure capacity without confounding it with a particular encoder's quality, we directly optimize unconstrained document and query vectors of a given dimension to realize a target pattern, using a pairwise margin objective, and report the fraction of queries for which every relevant document outranks every non-relevant one ("realizability"). This is the best case for *any* encoder of that dimension, so it upper-bounds real models and isolates the geometric limit; where we later want a smoother signal we report recall@k instead. The same primitive drives the retrieval probe (optimize $\mathbb{R}^{d_r}$ vectors against a relevance pattern) and the compression probe (a slot-memory read head over m codes of width d_c), which is what lets us compare the two walls on the same axis. All runs are seeded and multi-seed for headline claims, and we report compute alongside accuracy; experiments ran on single commodity GPUs (Kaggle T4/P100).

4 Two walls of the same geometry (C1)

The retrieval wall. On the adversarial "all-pairs" pattern, where every document pair is some query's gold set ($k = 2$), free-embedding realizability shows a sharp phase transition in d_r: for a corpus of $n_d = 40$ it jumps from 0.38 to 1.0 between $d_r = 4$ and $d_r = 8$ across three seeds with

Table 1: **The retrieval wall generalizes** across a second embedder, a second dataset, and optimal (PCA) truncation. We report recall@10 at truncated dimension d_r and the fraction of full-dimension recall retained at $d_r = 256$. In every case a modest d_r captures the bulk and the remaining dimensions add little.

Embedder / data (trunc.)	full d	d_r=16	32	64	128	256	full	@256/full
mxbai / SciFact (Matryoshka)	1024	0.33	0.57	–	0.79	0.83	0.87	95%
arctic-m / SciFact (Matryoshka)	768	0.18	0.39	0.59	0.75	0.82	0.85	97%
mxbai / FiQA (Matryoshka)	1024	0.10	0.24	0.42	0.53	0.60	0.65	92%
bge-large / SciFact (PCA)	1024	0.49	0.65	0.76	0.80	0.85	0.86	98.5%

Table 2: **The wall on canonical adversarial data** (LIMIT, Weller et al., 2025). Full set: 50k documents, 1000 queries, two gold per query. All three real embedder families collapse; the failure is a property of the single-vector paradigm, not a model quirk.

	LIMIT-full (50k docs)				LIMIT-small (46)
Embedder	R@2	R@10	R@100	R@1000	R@10
mxbai-embed-large	0.004	0.010	0.028	0.103	0.441
arctic-embed-m	0.009	0.024	0.082	0.254	0.619
bge-base	0.007	0.015	0.045	0.163	0.492

near-zero variance, and the critical dimension $d_r^\star$ moves outward as the corpus grows. The growth is real but slow: on a five-seed, two-pattern sweep (all-pairs and random 3-subsets) $d_r^\star$ moves only from 8 to 11 as n_d goes from 40 to 400, consistent with a sub-logarithmic law (a power fit gives exponent $b \approx 0.16 - 0.18$), and we note honestly that a tidy "$d_r^\star \propto d/n_d$" sub-critical collapse we initially conjectured does *not* hold ($R^2 = 0.18$); what survives, and is all the allocation law needs, is that a modest critical dimension exists and grows with content.

The wall is not a free-vector artifact. Truncating a Matryoshka encoder (Kusupati et al., 2022) and re-normalizing traces the same ramp-then-saturate curve on real data: `mxbai-embed-large` on SciFact climbs recall@10 from 0.09 to 0.87 and is statistically flat beyond $d_r = 512$, so the top half of a 1024-dimensional production embedding is wasted budget. Table 1 shows this replicates across settings that rule out the obvious objections: a *second* embedder family (`arctic-embed-m`, 768d) reaches 97% of full recall by $d_r = 256$; a *second* dataset (FiQA, a domain shift) reaches 92% by $d_r = 256$; and under *PCA* truncation—the optimal d_r-dimensional linear subspace, not a Matryoshka ordering—`bge-large` saturates hard at $d_r = 256$ (98.5% of full, the top 768 dimensions adding 0.013), which is the cleanest wall of the set and the one that kills the "your wall is just Matryoshka" reading.

Most tellingly, the wall appears on the field's own adversarial benchmark. On the full LIMIT set (Weller et al., 2025) (50k documents, two gold per query) three real embedders collapse to recall@100 of 0.028, 0.082, and 0.045 (Table 2); on the 46-document LIMIT-small, recall@10 is only $0.44 - 0.62$, i.e. returning ten of just forty-six documents still misses half the gold. That three different embedder families fail together, and consistently, is what makes this the wall of the single-vector *paradigm* rather than any one model's quirk.

The compression wall. The compression stage carries a wall of the same form. On controlled passages of n_f facts read from m soft codes, slot-memory realizability transitions at a critical code size $D_c^\star = m \cdot d_c \approx 2n_f$, growing roughly linearly with content, and the transition is softer than the retrieval one but unmistakable. The wall survives on a real encoder: a single 1024-dimensional `mxbai` embedding used as the compressed code holds about one fact, with a light probe recovering the queried value at $0.98, 0.53, 0.32, 0.24, 0.20$ for $n_f = 1, 2, 4, 8, 16$ (chance 0.125), and giving each fact its own slot restores capacity to $0.96 - 0.99$ once the per-slot width clears a floor. We also flag a scope boundary honestly: under a learned attention-read code, capacity is not shape-invariant—one fat token and many thin tokens both collapse while an interior slot count is best—but under real-encoder truncation only the thin-slot collapse remains, so the interior optimum is specific to the learned-code mechanism rather than a universal law. Two points guard against the objection that this is a probe artifact. First, the same wall holds on a second embedder family (`bge-large`,

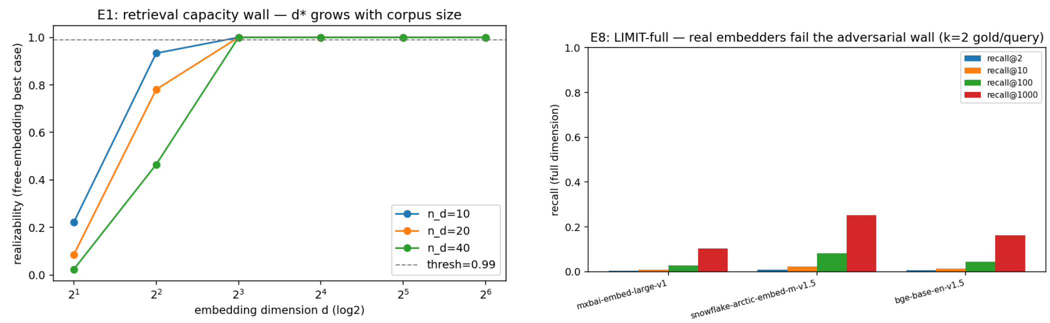

Figure 2: **C1.** Left: the free-embedding retrieval wall—realizability jumps sharply once d_r clears a critical value that grows with corpus size n_d. Right: on LIMIT, three real embedder families all fail the adversarial pattern at full dimension.

App. A), so it is not specific to one encoder. Second, and more directly, a *deployed* soft-token compressor shows exactly this wall in the wild: Belikova et al. (2026) detect "overflow" in the one-token `xRAG` (Cheng et al., 2024) code—a regime where the compressed token can no longer answer the query—which is precisely the capacity threshold we characterize, so our account explains an independently observed phenomenon rather than resting on the probe alone. The probe is best read as an *upper bound*: it measures the code's information ceiling with a light, reliably-trained reader, so a fully-trained generative soft-compressor can only do worse (indeed our own from-scratch attempt sits at chance without a large training budget, App. B).

5 COMPOSING THE TWO STAGES COMPOUNDS (C2)

Independent composition already bites. If the two stages fail on independent items, the pipeline recall is the product $p_R(d_r) \cdot p_C(D_c)$, so composing two sub-full stages must underperform either one at the full budget. On a synthetic corpus that shares a budget $B = d_r + D_c$ between the real retrieval and compression walls of Section 4, the best split at $B = 64$ reaches only 0.559 even though each stage alone exceeds 0.95 at that budget—a compounding gap of 0.389—and the gap closes to 0.001 by $B = 192$ once both stages clear their walls. The effect is largest exactly where deployments live, at tight budgets where neither stage is saturated.

The deviation from the product law is set by a hardness correlation. Independence is a baseline, not a law, so we model the two per-item success events with a Gaussian copula that holds the marginals fixed at the measured walls and varies the correlation ρ between which items are hard to retrieve and which are hard to compress. The limits are exact: $\rho = 0$ recovers the product $p_R p_C$; aligned hardness ($\rho > 0$) makes failures redundant and lifts the pipeline toward $\min(p_R, p_C)$; and misaligned hardness ($\rho < 0$) is the only regime that compounds *super*-multiplicatively, pushing the pipeline below $p_R p_C$ toward the Fréchet floor $\max(0, p_R + p_C - 1)$. Concretely, at a balanced 32:32 split the pipeline moves from 0.515 at $\rho = -0.5$ (below the independent 0.559) up to 0.687 at $\rho = +0.9$. So the sign of the compounding is an empirical question about real corpora, which prior work left open.

We measure ρ, and it is near zero. On a real pipeline—a truncated `mxbai` retriever and a real-encoder compressor over one shared corpus—we record, per query, a retrieval difficulty (gold-versus-best-distractor margin) and a compression difficulty (the reader's correct-versus-best-other logit margin), and correlate them. Across three balanced, sub-saturated operating points the mean ρ_ϕ is $+0.009$, i.e. essentially zero, and the copula prediction reproduces the observed pipeline recall almost exactly: observed $\{0.234, 0.443, 0.555\}$ against the product law $\{0.227, 0.439, 0.555\}$, a mean absolute error of 0.0035 (independent) or 0.0012 (with the measured ρ). The reading is that real RAG, at least on a benign corpus, lives in the multiplicative regime, so the pessimistic super-multiplicative case needs adversarial anti-correlation that we do not see by default—which is reassuring for allocation, since it means $p_R p_C$ is a near-exact objective to optimize.

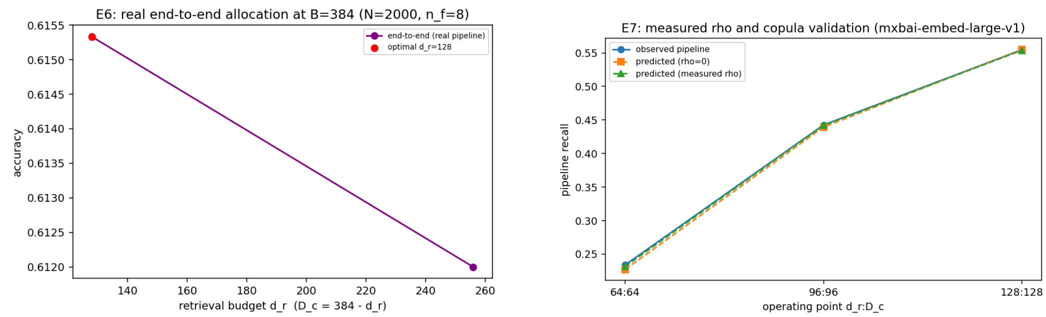

Figure 3: **C2–C3 on real models.** Left: end-to-end accuracy versus the retrieval share of a fixed budget—the optimum is interior, and the big-embedder/tiny-compressor corner is off it. Right: measured $\rho \approx 0$; the copula (product law) reproduces the observed pipeline recall at every operating point.

6 Budget-optimal allocation (C3)

Because the pipeline is (near-)multiplicative, the optimal split of a fixed budget follows a simple recipe: fund retrieval up to its wall $d_r^\star$, then pour the remainder into compression, since dimensions past the retrieval wall buy almost nothing while the compressor is still starved. On the synthetic composition this is exactly what we see—the optimal split is $48{:}(B-48)$ for all $B \geq 96$, with a balanced 32:32 preferred only when B is below twice the wall—and it holds end-to-end on real encoders. On one shared corpus with a real `mxbai` retriever and a real-encoder compressor, the best split at $B = 128$ is interior (64:64) and reaches 0.234 while the stages alone reach 0.63 and 0.89, a compounding gap of 0.392 that shrinks to 0.137 by $B = 256$ (Figure 3, left). The practical statement is blunt: the standard configuration of a large $d_r \approx 768-1024$ embedding with a tiny $m \approx 1-8$ compressor sits far off the frontier, since it spends heavily on embedding dimensions past the retrieval wall while the compressor overflows, and re-allocating the same budget buys accuracy for free. This is not just a frontier picture but a concrete intervention: on real QA (E9, below), moving the shared budget from the deployed big-retrieval/tiny-compression corner to the interior optimum raises answer F1 by $+0.032$ (1.5B reader) and $+0.035$ (7B) at $B{=}256$, at identical token cost. Combined with C1, which locates each wall cheaply by truncation, this is an actionable procedure: measure both walls, then split the budget so retrieval is funded to $d_r^\star$ and the rest goes to compression.

Confirmation on a real multi-hop QA benchmark (E9). The synthetic pipeline of E6 isolates the mechanism, but the claim that matters is whether it survives on an actual question-answering task with a real frozen reader. We pool every HotpotQA question's paragraphs into one shared corpus (a genuine retrieval problem, not one isolated per question), truncate a real embedder to d_r for retrieval, select sentences into a fixed token budget by D_c-truncated query similarity (the deployable, selective instantiation of compression used by RECOMP-style systems), and let a frozen `Qwen2.5-1.5B-Instruct` answer, scored by EM/F1 over three seeds. Both claims survive on real answer quality (Figure 4). The allocation curve is single-peaked with an *interior* optimum: F1 rises to a peak near $d_r \approx 80-96$ and then declines along a long tail out to $d_r{=}224$, so past the optimum more retrieval budget *hurts*, because it starves compression— exactly the shape C3 predicts. And composition compounds at the tight budget: at $B = 128$ the best split reaches F1$=$ 0.335 against standalone references of 0.39 (retrieval) and 0.56 (compression), a compounding gap of $\mathbf{0.048 \pm 0.007}$ F1 with all three seeds positive; the gap shrinks to 0.009 ± 0.015 (null) by $B = 256$ once both stages clear their walls, the same shrinking-with-budget pattern as the synthetic result and consistent with the measured $\rho \approx 0$. The precise optimum location is noisy across seeds ($d_r^\star$ in 64–160), but its interior character and the tight-budget compounding gap are robust—the real-QA, EM/F1 counterpart of E3 and E6. The gap is also *reader-scale robust*: across a $5\times$ range of reader size (`Qwen2.5` at 1.5B, 3B, and 7B, three seeds each) the tight-budget ($B{=}128$) gap stays flat at $0.048/0.058/0.058$ F1 and the interior optimum holds at every scale—it does not vanish or reverse with reader capability. This is the signature of an *information* bottleneck rather than a reasoning one—a more capable reader extracts more from good context, so the penalty for the evidence the

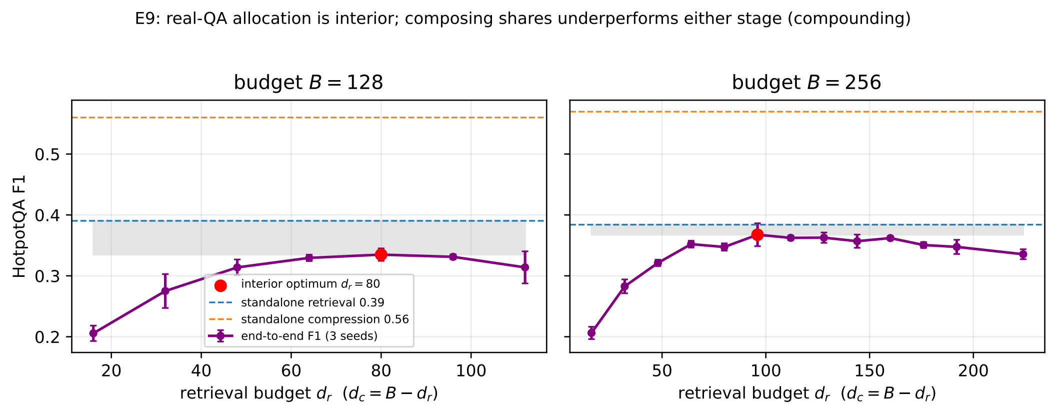

Figure 4: **C2–C3 on real multi-hop QA** (HotpotQA, `Qwen2.5-1.5B` reader, EM/F1, three seeds). End-to-end F1 versus the retrieval share of a fixed budget. The curve is single-peaked with an interior optimum (red), and past it more retrieval budget hurts (it starves compression). The shaded band is the compounding gap—the best split underperforms the weaker standalone stage—which is positive at the tight budget $B{=}128$ (0.048 ± 0.007 F1) and vanishes by $B{=}256$.

two stages discard persists (and at the looser budget grows: $0.009{\to}0.036{\to}0.027$), unlike a context-cleanup heuristic whose benefit a stronger reader absorbs. It is also dataset-general: on a second multi-hop benchmark (2WikiMultihopQA, three seeds) the tight-budget gap is 0.048 ± 0.014 F1 with the same single-peaked interior optimum.

7 When does multi-view help? The escape and its scope (C4)

A natural way to beat a single-vector wall is to stop using a single vector. We test a facet-lens: K low-rank lenses, each specialized to one semantic facet (entities, numerics, relations, negation, $\ldots$), with a lightweight router sending each query to its facet's lens, so every doc gets K small codes instead of one large one at equal total budget. In the free-vector worst case this escapes the wall cleanly. On a facet-structured pattern, routed facet-lenses realize the relevance structure at critical budget $d_r^\star = 12$ against a single vector's 24—half the budget—and, tellingly, a generic K-view MaxSim scheme with no routing is *worse* than a single vector ($d_r^\star = 48$), so the lever is specialization plus routing, not multi-vector per se. This distinguishes the facet-lens from late-interaction multi-vector retrieval (Khattab & Zaharia, 2020), where the gain comes from many token vectors rather than from routing to a facet.

We then report the honest negative: the escape does *not* transfer to a strong real encoder. Learning low-rank lenses over frozen `mxbai` embeddings of a facet-structured corpus (ten facets), a single *learned* projection ties the routed lenses—both reach mean average precision 0.9 at budget 40—and the single projection actually leads at tight budgets (0.735 versus 0.581 at budget 20), while generic multiview is far behind (budget 320). The mechanism is instructive: a powerful pretrained encoder has already spent representational capacity making the facets close to linearly separable, so a single learned projection can unmix them and routing adds little, whereas the free-vector escape was a worst-case sign-rank phenomenon in which no single set of vectors could satisfy all facet constraints at once. The consequence for practice is clean and, we think, more useful than a method that only wins in the worst case: on real encoders the lever is allocation (C3), not multi-view, and multi-view is not a free lunch. We therefore keep C4 as a scoped "when does it help" result rather than the paper's headline, which protects the empirical core (C1–C3) from an overclaim.

8 Discussion and limitations

The account is deliberately a capacity characterization rather than a closed-form theorem. The free-vector measure is a best case, so real encoders can only do worse, which is why the wall shows up so cleanly on truncated production embeddings and on LIMIT; but it also means our $c_{\text{eff}} \leq \min(\cdot)$

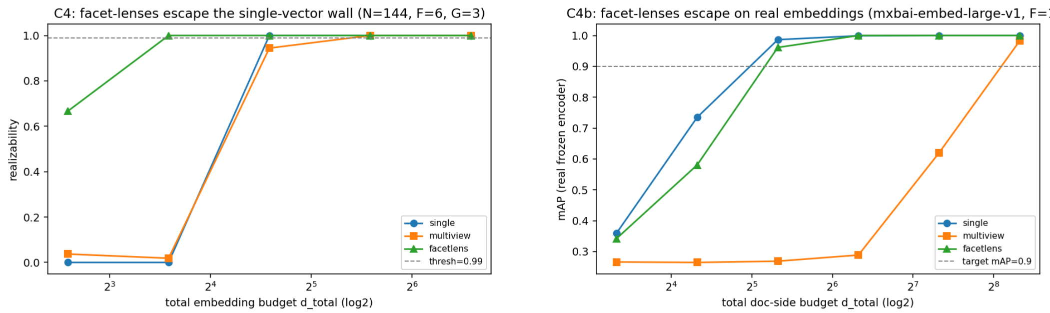

Figure 5: **C4 and its scope.** Left (free vectors): routed facet-lenses escape the wall at half the budget of a single vector, while generic multiview is worse than single. Right (frozen real encoder): the escape vanishes—a single learned projection ties the routed lenses and leads at tight budgets—because a strong encoder has already linearized the facets.

statement is a proposition about the realizable set backed by a direct empirical measurement of that set, not a bound with constants. Several boundaries are worth stating plainly. The compounding experiments use controlled single-task corpora so that we can hold the marginals fixed and vary ρ; the measured $\rho \approx 0$ is on a benign corpus, and whether adversarial inputs can drive the *joint* ρ negative—LIMIT shows retrieval can be driven near zero, but the joint hardness correlation under such pressure is untested—is the one genuinely open question we would flag, since a negative ρ is where the pessimistic super-multiplicative regime lives. The compression capacity is isolated with a light probe rather than a fully-trained generative soft-token compressor, which in our hands stays at chance without a large training budget; the probe cleanly establishes the capacity ceiling but does not model the end-to-end training dynamics of methods like Cheng et al. (2024); Ge et al. (2023). Finally, C4's escape is scoped out of the strong-real-encoder regime, so we do not claim multi-view as a general remedy; we claim, more narrowly and we think more usefully, that specialization-plus-routing is what would be needed for an escape and that a strong encoder makes it redundant.

9 Conclusion

We treated the retriever and the compressor of efficient RAG as two fixed-dimensional vector bottlenecks of the same geometric form, and showed that composing them compounds: each stage has a capacity wall that grows with content and that appears on real encoders, on a second dataset, under optimal truncation, and on the adversarial LIMIT set; the composition underperforms either stage, with the deviation from the product law set by a hardness correlation that we measured to be near zero, which places real RAG in a multiplicative regime; and the resulting allocation is interior, so the deployed habit of a large embedding with a tiny compressor is off the frontier and can be improved at equal cost. We also tested the obvious multi-view escape and reported honestly that it does not transfer to strong real encoders, which points the practical lever back at allocation. The picture that emerges is a full arc—a diagnosis of the problem, an actionable allocation for it, and a careful account of when a fancier representation would and would not help—and we hope it reframes "efficient RAG" as a budget-allocation problem across two coupled bottlenecks rather than two tricks tuned in isolation.

Reproducibility statement

Every experiment is driven by a committed YAML configuration and a fixed seed, headline claims use at least three seeds with reported variance, and each run emits a machine-readable results file and a human-readable summary. The capacity primitives, synthetic-corpus generators, real-encoder truncation and probe code, the copula composition, and the facet-lens are all in the released source, and the figures and tables here are produced directly from the committed outputs. Real-model experiments use publicly available encoders (`mxbai-embed-large-v1`, `arctic-embed-m-v1.5`, `bge-large-en-v1.5`, `all-MiniLM-L6-v2`) and datasets (SciFact, FiQA, LIMIT) and run on single commodity GPUs.

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A Summary of experiments

Table 3 collects every experiment behind the claims in the main text, with the setting and the headline number, so the results can be traced to a single row. Each corresponds to a committed configuration and a fixed seed; headline claims use at least three seeds.

B Implementation details

Free-embedding capacity primitive. For a target relevance pattern $A \in \{0,1\}^{M\times N}$ we optimize unconstrained document vectors $X \in \mathbb{R}^{N\times d}$ and query vectors $Q \in \mathbb{R}^{M\times d}$ with Adam on a pairwise margin loss that, per query, pushes the smallest relevant score above the largest non-relevant score by a margin; realizability is the fraction of queries for which this holds strictly. Because X, Q are free, this upper-bounds any encoder of dimension d, so a wall here is a wall for every encoder. The all-pairs pattern sets every document pair as one query's gold set (k=2); random-k draws n_q distinct k-subsets.

Compression probe. A passage of n_f facts is written as m codes of width d_c; a query attends over the m codes with a small read head and decodes the queried value with cross-entropy. On real encoders the codes are truncated frozen embeddings, and only the light read head is trained, which converges reliably and generalizes to held-out passages, so it measures the code's capacity rather than the trainability of a full soft-token compressor (which, in our hands, stays at chance without a large training budget).

Copula composition and measured ρ. We place a Gaussian copula on the two binary per-item success events with the marginals fixed at the measured stage recalls and correlation ρ, and estimate pipeline recall by Monte-Carlo; the limits $\rho \in \{-1, 0, +1\}$ recover the Fréchet lower bound, the product, and the minimum. To measure ρ on a real pipeline we take the phi coefficient of the two binary successes and the Pearson correlation of two continuous difficulty margins (retrieval gold-minus-best-distractor; compression correct-minus-best-other logit).

Facet-lens. Each lens is a learned linear map $\mathbb{R}^D \to \mathbb{R}^{d/K}$ applied to frozen document and query embeddings, scored by $\langle W_q^f q, W_d^f d\rangle$; the router sends a query of facet f to lens f, so the doc-side budget d is split into K equal codes. Single is one full-width lens; generic multiview keeps K lenses but scores by the max over views without routing. We train all conditions with the same objective and budget and report mean average precision.

Table 3: All experiments, their setting, and the headline result. "Free" denotes the free-embedding capacity primitive of Section 3; "real" denotes a frozen production encoder. Budgets are embedding/code dimensions.

ID	Kind	Setting	Headline result
E1/E1c	free	all-pairs & random-k patterns; sweep d_r, n_d; 5 seeds	sharp wall; $d_r^\star$ grows $8 \to 11$ as n_d $40 \to 400$; "d/n_d" law refuted (R^2=0.18)
E2	free	slot-memory read; sweep D_c, n_f	compression wall $D_c^\star \approx 2n_f$
E4	real	mxbai / SciFact; Matryoshka trunc.	recall@10 $0.09 \to 0.87$; flat past d_r=512
E4b	real	arctic-m / SciFact; Matryoshka	97% of full recall by d_r=256
E4c	real	mxbai / FiQA (domain shift)	92% of full recall by d_r=256
E4d	real	bge-large / SciFact; **PCA** trunc.	saturates at d_r=256 (98.5%)
E5/E5b	real	mxbai code; light read probe	one vector $\approx$ one fact ($0.98 \to 0.20$); multi-token recovers to 0.99
E5c	real	**bge-large** code; same sweeps	compression wall replicates ($D_c^\star$ grows with n_f); F6 thin-slot scope holds
E8	real	LIMIT; 3 embedders; 50k / 46 docs	recall@100 = $2.8/8.2/4.5\%$ (full)
E3	free	shared budget B=d_r+D_c	compounding gap 0.389 at B=64
E3b	free	Gaussian copula over stage successes	sign of gap set by ρ; Fréchet-bounded
E7	real	mxbai retriever + compressor; per-item margins	$\rho_\phi = +0.009$; product law exact to $\sim$0.001
E6	real	mxbai; one corpus; N=2000	gap 0.392 at B=128; optimum interior 64:64
E9	real	HotpotQA; Qwen2.5-1.5B; EM/F1; 3 seeds	interior optimum; gap 0.048 ± 0.007 F1 at B=128, null at 256
E9c/E9e	real	HotpotQA; Qwen2.5-**3B/7B**; 3 seeds	reader-robust: B=128 gap $0.058/0.058$ (flat over 1.5B/3B/7B, no reversal)
E9d	real	**2Wiki**; Qwen2.5-1.5B; 3 seeds	cross-dataset: gap 0.048 ± 0.014 at B=128; same interior optimum
C4	free	facet pattern; single / multiview / routed	routed $d_r^\star$=12 vs single 24; multiview 48
C4b	real	frozen mxbai; 10 facets; learned lenses	single ties routed ($d_r^\star$=40); no transfer

Compute. All runs use a single commodity GPU (Kaggle T4 or P100). Real-encoder runs encode the corpus once and cache it; wall-clock ranges from seconds (LIMIT-small; the free-vector sweeps) to about twenty-six minutes for the 50k-document LIMIT-full sweep across three embedders.